MUHAMMAD INUWA MUHAMMAD

AI / Machine Learning Engineer • Deep Learning • Agentic AI Systems • Computer Vision

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PROFESSIONAL SUMMARY

Innovative AI/ML Engineer and Mechatronics Engineering student at the Federal University of Technology Minna, with a proven track record of building and deploying intelligent systems across medical imaging, autonomous robotics, NLP, computer vision, and agentic AI. I architect end-to-end ML pipelines — from raw data and model design through to deployment and user impact — and have delivered projects serving real users in healthcare, finance, transport, and productivity. Recognized nationally as First Runner-Up at the RAIN x Meta AI Developer Academy Nigeria Hackathon (2025) and selected as a UN Millennium Fellow. I bring both research depth and engineering discipline to every problem.

PROJECTS

Physics-Informed MRI Reconstruction for Low-Field MRI, *PyTorch • E2E-VarNet • Medical Imaging • Deep Learning*, *2025*

- **Motivation:** addressed the clinical challenge of low-field MRI scanners producing noisy, low-resolution images with long scan times — particularly impactful for resource-limited settings in Nigeria and emerging markets.
- Implemented and extended the End-to-End Variational Network (E2E-VarNet) as a baseline unrolled reconstruction model, incorporating physics-informed priors (MRI forward model) into the network's iterative refinement loop.
- Achieved a Structural Similarity Index (SSIM) of 0.86 on reconstructed images — significantly improving diagnostic image quality while enabling a reduction in scan acquisition time.
- Developed a full data preprocessing pipeline for k-space undersampled MRI data, applying retrospective undersampling masks to simulate accelerated scan scenarios.
- Evaluated reconstruction quality across multiple acceleration factors using SSIM, PSNR, and NMSE metrics to rigorously validate model performance.

Road Abnormalities Detector, *PyTorch • TensorFlow • CNN • Deep Learning • OpenCV*, *2025*

- Led a 13-member multidisciplinary team to build a deep learning model detecting road cracks, potholes, and bumps from images — achieving high detection accuracy across diverse road conditions.
- Managed the full ML pipeline: data collection, annotation, preprocessing, model training, hyperparameter tuning, and evaluation using precision, recall, and F1 metrics.
- Presented as a viable smart infrastructure AI tool, contributing to road safety and urban planning use cases.

Smart Motion Bin — Autonomous Office Robot, *JavaScript • Firebase Realtime DB • IoT • Embedded Systems*, *2026*

- Architected and built an autonomous indoor robot navigating to user workstations on command, bridging cloud-based AI control with physical hardware.
- Engineered a real-time web command center using JavaScript and Firebase for low-latency bidirectional control, with live telemetry for IR sensors, obstacle radar, and battery monitoring.
- Demonstrated full-stack hardware-software integration: embedded systems, cloud pipelines, and autonomous spatial navigation.

Multimodal AI Loan Advisor, *LLaMA 3.1 • Whisper • Tesseract OCR • PyPDF2 • gTTS • Pandas*, *2025*

- Built a financial literacy AI system for rural farmers and students, processing loan documents in 8 Nigerian languages using a multimodal pipeline (OCR, ASR, TTS, and LLM).
- Engineered a risk-assessment engine using Meta LLaMA 3.1 to auto-detect hidden fees and classify contract risk levels, paired with a Pandas-driven loan amortization calculator.
- Demonstrated responsible AI deployment for underserved communities — handling multilingual, multimodal inputs in low-resource environments.

ProDOS — Agentic AI Productivity Bot, *Telegram API • Agentic AI • Python*, *2026*

- Designed and deployed an autonomous productivity AI bot on Telegram, scaling to 65+ active student users through iterative product improvement and user feedback loops.
- Integrated automated progress logging, peer competition features, and social campaign automation — demonstrating real-world agentic AI system design.

AI Booking Automation System, *Telegram Bot API • Python • Payment Integration*, *2024 – 2025*

- Engineered a Telegram-based AI booking system for Togo Mobility Limited, automating passenger registration and payment verification via a 7-digit secure code — reducing manual booking time by 60%.
- Onboarded 20+ local riders through bilingual (English/Hausa) outreach, expanding platform capacity in previously untapped local markets.

WORK EXPERIENCE

AI Developer & Customer Support | **Togo Mobility Limited** | *2024 – 2025*

- Delivered a production AI automation system that reduced booking processing time by 60% and onboarded 20+ riders.
- Provided bilingual (English/Hausa) technical and customer support, bridging communication gaps for non-English-speaking users.

Community Manager & Customer Support | **Jekaeat** | *2023*

- Grew community engagement by 40% through strategic content curation and direct buyer-brand relationship management.
- Resolved 95%+ of customer inquiries within SLA in a high-volume, fast-paced environment — demonstrating precision under pressure.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, LightGBM, CatBoost, Optuna, Hugging Face
Medical Imaging & CV: E2E-VarNet, MRI reconstruction, k-space processing, CNN, OpenCV, SSIM/PSNR evaluation
NLP & Multimodal AI: LLMs (LLaMA 3.1), Transformers, Whisper ASR, Tesseract OCR, gTTS, RAG, Prompt Engineering
Agentic AI: Multi-agent orchestration, tool-use pipelines, Telegram Bot API, autonomous task execution
Data Engineering: Pandas, NumPy, ETL pipelines, data preprocessing, annotation, quality validation
Cloud & IoT: Firebase Realtime Database, REST APIs, CI/CD pipelines, embedded systems (Arduino, ESP32)
Languages: Python (expert), JavaScript, MATLAB, SQL
Human Languages: English — Native/Bilingual | Hausa — Fluent

EDUCATION

Bachelor of Engineering (B.Eng) — Mechatronics Engineering
Federal University of Technology Minna, Nigeria | 400 Level (In Progress)

ACHIEVEMENTS & RECOGNITION

- First Runner-Up — RAIN x Meta AI Developer Academy Nigeria Hackathon (2025): Nationally recognized for applied AI innovation and engineering excellence.
- Millennium Fellow — United Nations Academic Impact & Millennium Campus Network (2025): Selected globally for leadership, innovation, and social impact contributions.
- Scholarship Recipient — Halima Ibrahim Babangida Foundation (2024).